Hello World

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First document. This is my hello World of Latex. The first document is usually easy to start and execute.

Some of the **bold text are important** in notes making for reading note.

Some of the greatest *discoveries* in science were made by accident.

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This is a normal paragraph. LaTeX automatically indents the first line of each paragraph except after section headings. The indentation size is controlled by `\parindent`.

This is another paragraph. You can see the first line is indented automatically. To create a new paragraph, just leave a blank line in your source code.

This is still the same paragraph. See how it continues without indentation?

As you can see in figure 1 above 1, the image is AI generated . This image can be generated using AI tools.

- This is list 1.
- This is list 2.

1. This is first part of ordered list.
2. This is the second part of ordered list.

- In physics, the mass-energy equivalence is stated by the equation $E = mc^2$, discovered in 1905 by Albert Einstein.

Figure 1: Fig 1.0, AI Puppy
content...

- Further mathematics formula. The formula $E = mc^2$ is famous.

$$\int_0^1 x^2 dx = \frac{1}{3}$$

$$a^2 + b^2 = c^2 \tag{1}$$

Subscripts in math mode are written as a_b and superscripts are written as a^b. These can be combined and nested to write expressions such as

$$T^{i_1 i_2 \dots i_p}_{j_1 j_2 \dots j_q} = T(x^{i_1}, \dots, x^{i_p}, e_{j_1}, \dots, e_{j_q})$$

We write integrals using $\int$ and fractions using $\frac{a}{b}$. Limits are placed on integrals using superscripts and subscripts:

$$\int_0^1 \frac{dx}{e^x} = \frac{e-1}{e}$$

Lower case Greek letters are written as ω δ etc. while upper case Greek letters are written as Ω Δ.

Mathematical operators are prefixed with a backslash as $\sin(\beta)$, $\cos(\alpha)$, $\log(x)$ etc.

1 First example

The well-known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents, meaning the next equation has no integer solutions for $n > 2$:

$$x^n + y^n = z^n$$

2 Second example

This is a simple math expression $\sqrt{x^2+1}$ inside text. And this is also the same: $\sqrt{x^2+1}$ but by using another command.

This is a simple math expression without numbering

$$\sqrt{x^2+1}$$

separated from text.

This is also the same:

$$\sqrt{x^2+1}$$

...and this:

$$\sqrt{x^2+1}$$